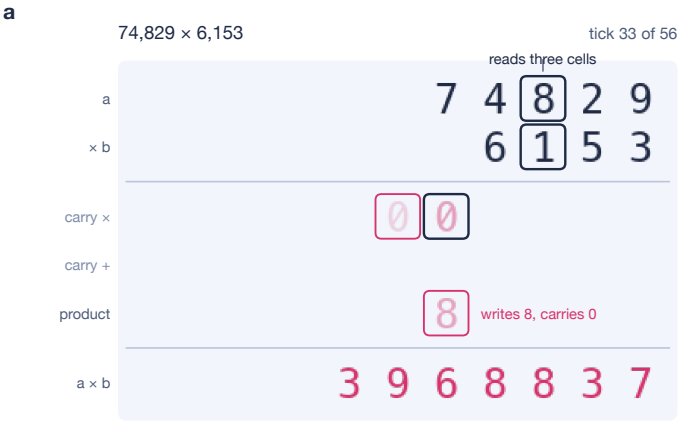
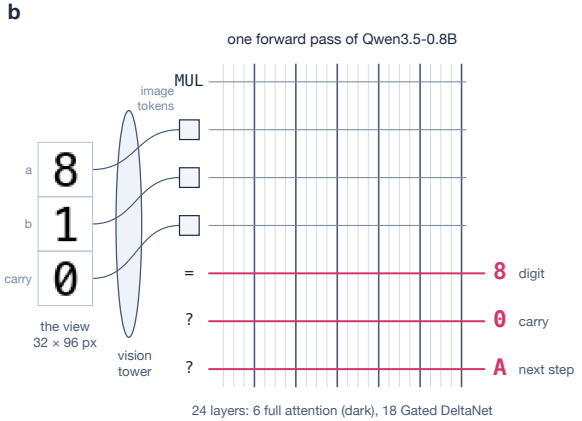
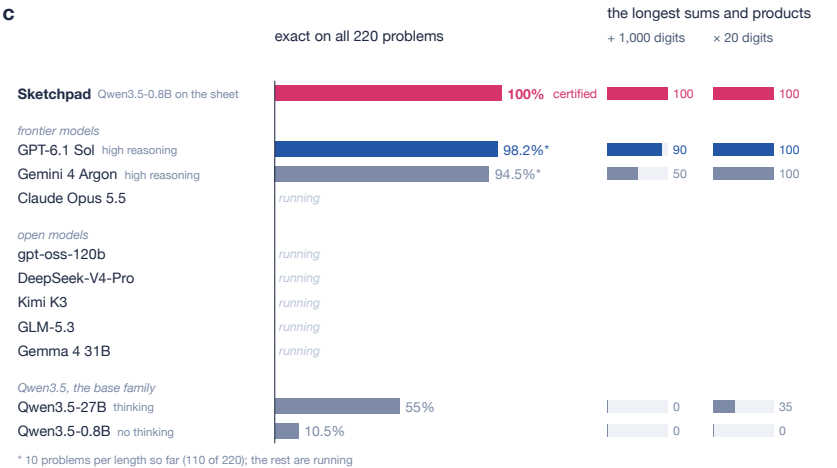
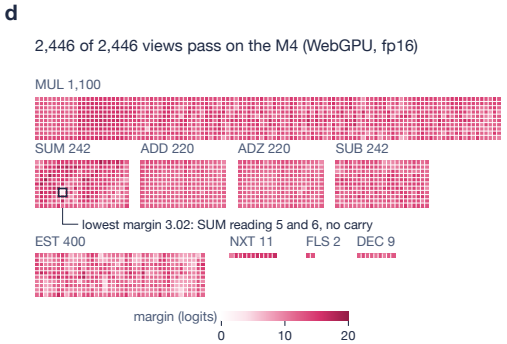

a
74,829 × 6,153
tick 33 of 56
reads three cells
a
× b
7 4 8 2 9
6 1 5 3
carry ×
carry +
product
writes 8, carries 0
a × b
3 9 6 8 8 3 7


b
one forward pass of Qwen3.5-0.8B
image tokens
MUL
a
b
carry
8
1
0
the view
32 × 96 px
vision tower
=
?
?
8 digit
0 carry
A next step
24 layers: 6 full attention (dark), 18 Gated DeltaNet


c
exact on all 220 problems
the longest sums and products
+ 1,000 digits
× 20 digits
Sketchpad Qwen3.5-0.8B on the sheet
100% certified
100
100
frontier models
GPT-6.1 Sol high reasoning
98.2%*
90
100
Gemini 4 Argon high reasoning
94.5%*
50
100
Claude Opus 5.5
running
open models
gpt-oss-120b
running
DeepSeek-V4-Pro
running
Kimi K3
running
GLM-5.3
running
Gemma 4 31B
running
Qwen3.5, the base family
Qwen3.5-27B thinking
55%
0
35
Qwen3.5-0.8B no thinking
10.5%
0
0
* 10 problems per length so far (110 of 220); the rest are running


d
2,446 of 2,446 views pass on the M4 (WebGPU, fp16)
MUL 1,100
SUM 242
ADD 220
ADZ 220
SUB 242
lowest margin 3.02: SUM reading 5 and 6, no carry
EST 400
NXT 11
FLS 2
DEC 9
margin (logits)
0
10
20